

Reinforcement Learning for Health Monitoring Using Wearable Sensor Data

Abstract:

This research investigates the use of Reinforcement Learning (RL) in health monitoring through data obtained from wearable sensors. As wearable technology becomes increasingly common, there is a significant opportunity to leverage RL for the development of advanced health management systems. A review of current literature indicates that although RL has been effectively utilized for promoting behavior change and optimizing energy-efficient data transmission, there has been limited focus on real-time detection of health anomalies using straightforward and adaptable RL methods. This project intends to address this gap by developing a lightweight Q-learning-based system aimed at identifying anomalies in wearable sensor data. By prioritizing simplicity, interpretability, and suitability for resource-limited settings, this research aspires to offer a practical solution for RL-enhanced health monitoring, facilitating early intervention and enhancing patient outcomes.

Introduction:

The rapid advancement of wearable technologies has transformed how individuals monitor their health, enabling continuous tracking of vital signs such as heart rate, physical activity, sleep patterns, and more. These devices generate large volumes of real-time physiological data, creating opportunities for intelligent systems to detect early signs of health anomalies and intervene promptly. However, traditional rule-based systems often struggle with the variability and complexity of human health data, making them less adaptable in dynamic environments.

Reinforcement Learning (RL), a subfield of machine learning where agents learn optimal actions through interactions with an environment, has shown promising applications in personalized healthcare. RL models can learn to make context-aware decisions, making them ideal for systems that must adapt to individual users' behaviors and physiological responses. Previous studies have explored RL for promoting physical activity, optimizing energy usage in wearable sensor networks, and detecting anomalies in general time-series data. However, these works either focus on behavior modification or require computationally intensive models that are unsuitable for lightweight or student-level projects.

This project aims to investigate the use of RL—specifically Q-learning—for detecting anomalies in health data collected from wearable sensors. The objective is to build a simple, adaptable model that can operate in low-resource environments, making it feasible for early-stage research and practical deployment. By focusing on foundational RL techniques, this study bridges the gap between theoretical reinforcement learning and real-world health monitoring applications.

This review discusses four relevant studies and identifies key research gaps that this project seeks to address.

1. Reinforcement Learning for Encouraging Physical Activity in Diabetes Patients (Yom-Tov et al., 2016)

This study applied Q-learning to encourage diabetes patients to walk more, using personalized goal-setting. The agent adjusted recommendations based on previous user responses, effectively promoting activity.

Gap: Focused on behavior motivation rather than detecting health anomalies from physiological data.

2. A Health Monitoring System Using Q-Learning for WBAN (Academia.edu, 2024)

This work used Q-learning to create an energy-efficient data transmission system in Wireless Body Area Networks (WBANs). It minimized sensor communication to save battery while maintaining monitoring.

Gap: Prioritized transmission efficiency but did not analyze or interpret the health data itself.

3. RLAD: Time Series Anomaly Detection through Reinforcement Learning and Active Learning

This paper introduces a reinforcement learning-based anomaly detection approach for time-series data, using active learning to adaptively detect anomalies in the data.

Gap: While this paper demonstrates the use of reinforcement learning and active learning for anomaly detection in time-series data, it does not specifically address health-related anomalies from wearable sensor data. The focus is on general anomaly detection in time-series data, which may not fully capture the nuances of physiological signals or health-related events that

could be detected by wearable sensors in real-time.

4. Deep Reinforcement Learning Empowered Activity-Aware Dynamic Health Monitoring Systems (Ye et al., 2024)

This study introduced the DActAHM framework, using Deep RL and SlowFast models to dynamically adjust health monitoring based on detected user activities via video data. It improved both system performance and energy efficiency.

Gap: Relies on complex models and video-based input, making it difficult to implement in lightweight or resource-limited environments — less suitable for simple RL exploration or student projects.

Proposed Idea:

In response to the identified gaps, this project proposes a Q-learning-based anomaly detection system for wearable health data. The system will be trained to recognize deviations in physiological signals (e.g., heart rate, movement patterns) that may indicate stress, fatigue, or abnormal health conditions. Unlike existing work that relies on complex deep learning models or focuses solely on behavioral nudging, this project emphasizes simplicity, adaptability, and real-time responsiveness. The goal is to create a lightweight prototype suitable for limited-resource environments while retaining meaningful health insight.

References:

. Reinforcement Learning for Encouraging Physical Activity in Diabetes Patients (Yom-Tov et al., 2016)

Link : <https://arxiv.org/abs/1605.04070>

. A Health Monitoring System Using Q-Learning for WBAN (Academia.edu, 2024)

Link:

https://www.academia.edu/115088287/A_Health_Monitoring_System_using_Reinforcement_Learning_based_Intellectual_and_Energy_Efficient_Transmission_for_WBAN_Model

. RLAD: Time Series Anomaly Detection through Reinforcement Learning and Active Learning

Link: <https://arxiv.org/abs/2104.00543>

. Deep Reinforcement Learning Empowered Activity-Aware Dynamic Health Monitoring Systems
(Ye et al., 2024)

Link: <https://arxiv.org/abs/2401.10794>

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